

ROBORACER GRAND PRIX 2026

Vehicle Technical Document

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1. Overall Vehicle Overview

1.1 Vehicle Design Concept

This autonomous racing vehicle is developed strictly following all hardware and software regulations of ROBORACER GRAND PRIX 2026. The vehicle adopts mature commercial 1/10 RC car chassis as the carrier, equipped with single 2D LiDAR for environment perception, NVIDIA Jetson Orin Nano as the only on-board computing unit, and single-motor independent power drive system. All computing tasks including perception, localization, path planning and vehicle control are completed **locally on the vehicle** without any external data transmission or remote human intervention during official races.

The whole vehicle is optimized for the smooth concrete track with urethane coating in the competition venue, adapting to low road friction coefficient and indoor scenarios.

1.2 Overall Compliance Summary

All core configurations of the vehicle fully comply with the 2026 ROBORACER competition technical constraints, including chassis dimension limits, maximum 4S LiPo battery limit, single LiDAR restriction, single power motor restriction, forbidden VIO camera restriction and exclusive single on-board computing unit requirement. No hardware with competitive advantages or rule-violating sensors is adopted.

2. Chassis Platform Configuration & Dimension Compliance Verification

2.1 Selected Chassis Information

Chassis Platform: Traxxas Slash 4X4

Model Part Number: #68086-4

Scale: 1/10 four-wheel drive short-course truck chassis

Chassis Material & Structure: Molded grooved chassis made of nylon composite material, with excellent shock resistance and structural stability adapting to indoor racing track bumps

Drive Mode: Shaft-driven full-time 4WD direct drive system

Steering Structure: Dual bell crank steering mechanism with integrated servo protector, ensuring stable steering accuracy during high-speed racing

2.2 Detailed Chassis Physical Parameters

Parameter Item	Specific Value

Overall Vehicle Length	568 mm
Front Wheel Track	296 mm
Rear Wheel Track	296 mm
Overall Vehicle Height	193 mm
Wheelbase	324 mm
Ground Clearance	72 mm
Curb Weight	2.64 kg
Tire Diameter	109.5 mm
Battery Compartment Size	165 mm × 49.5 mm × 23 mm, supporting extended battery compartment kit

2.3 Dimension Compliance Verification

Official competition dimension limits: Width: 238 mm ~ 341 mm; Length: 454 mm ~ 654 mm.

- Vehicle front/rear wheel track: 296 mm, located within the legal width range
- Vehicle overall length: 568 mm, located within the legal length range

Conclusion: The chassis dimension fully meets ROBORACER official requirements, and the dimension deviation is within 15% of standard Traxxas chassis size, complying with chassis access regulations.

3. On-board Battery Configuration & Power Compliance Verification

3.1 Battery Overall Configuration

The vehicle is equipped with two independent LiPo lithium polymer batteries for power supply. According to competition rules, the maximum single battery cell limit is 4S, and the total series cell count of all batteries shall not exceed 4S. The two batteries work independently without series-parallel connection for power output.

3.2 Detailed Battery Technical Parameters

Battery 1: 3S LiPo Power Battery

Battery Type: 3S Lithium Polymer Battery

Nominal Voltage: 11.1 V

Capacity: 5200 mAh

Discharge Rate: 40C

External Dimension: 28.5 mm × 43 mm × 135 mm

Function: Power supply for electronic speed controller (ESC) and driving motor

Battery 2: 4S LiPo Power Battery

Battery Type: 4S Lithium Polymer Battery

Nominal Voltage: 14.8 V

Capacity: 2200 mAh

Discharge Rate: 40C

External Dimension: 105 mm × 34 mm × 34 mm

Weight: 255 g

Interface Type: XT60

Function: Power supply for on-board main computing unit and all vehicle sensors (LiDAR)

3.3 Battery Compliance Verification

The highest series specification of single battery pack is 4S LiPo, which does not exceed the official maximum 4S battery limit. No series/parallel connection between two batteries, no over-limit battery combination. All battery configurations meet competition power supply regulations.

4. Perception Sensor Configuration & Compliance Verification

4.1 LiDAR Configuration (Single LiDAR Only)

In accordance with competition rule that only one LiDAR is allowed, the vehicle is equipped with **one single 2D scanning LiDAR**, no extra laser radar sensors are installed.

Brand: Hokuyo

Model: UST-10LX

Sensor Type: 2D scanning ranging LiDAR

Communication Interface: Ethernet (100BASE-TX)

Installation Quantity: 1 unit

Core Technical Parameters of Hokuyo UST-10LX

Parameter Item	Value
Ranging Range (High-reflection target)	0.06 m - 10 m
Ranging Range (10% low-reflection target)	0.06 m - 4 m
Maximum Detection Distance	30 m
Field of View	270°
Scanning Period	25 ms/scan
Motor Rotation Speed	2400 rpm
Angular Resolution	0.25°
Measurement Accuracy	±40 mm
Laser Safety Level	Class 1, 905 nm semiconductor laser
Protection Grade	IP65
Sensor Weight	130 g (excluding cable)

4.2 Camera Configuration

No camera module is equipped on this vehicle. The whole perception system only relies on LiDAR for environment mapping, obstacle detection and vehicle localization. No camera with onboard image processing function or VIO visual inertial odometry output function is adopted, which fully complies with camera usage prohibitions of the competition.

4.3 Perception Sensor Compliance Conclusion

Only one LiDAR is installed, no prohibited intelligent camera is equipped. All perception configurations fully meet ROBORACER sensor usage rules.

5. Power Drive System Configuration & Compliance Verification

5.1 Driving Motor Parameters (Single Motor Only)

Following the single motor power system rule, the vehicle adopts one independent brushless driving motor without any auxiliary driving motor.

Motor Brand: Traxxas

Motor Model: Velineon 3500

Motor Type: Brushless DC motor

Motor Quantity: 1 unit

Original Overall Gear Ratio: 11.82

Gear Pitch: 32 pitch

Adaptable Battery Specification: Native support for 3S LiPo battery operation

5.2 Electronic Speed Controller (ESC) Parameters

ESC Brand: XXD (Xin Xida)

ESC Model: HW30A

Type: Brushless aeromodelling electronic speed controller

Rated Working Current: 30 A

Quantity: 1 unit

Interface Definition: Three-phase output for brushless motor, two-wire DC battery input

5.3 Power System Compliance Conclusion

The vehicle adopts strictly single motor + single ESC power configuration, no additional power driving units. It fully meets the competition requirement of single motor unit for powertrain system.

6. On-board Computing Platform Configuration

In accordance with the rule of **only one main computing unit allowed**, the vehicle is equipped with a single NVIDIA Jetson Orin Nano 8GB development kit as the exclusive on-board computer, undertaking all autonomous driving computing tasks locally.

Product Model: MAKEROBOT CLB version NVIDIA Jetson Orin Nano 8GB

GPU Architecture: NVIDIA Ampere

CUDA Core: 1024

Tensor Core: 32

CPU: 6-core Arm Cortex-A78AE v8.2 64-bit processor

On-board Memory: 8 GB 128-bit LPDDR5

Memory Bandwidth: 68 GB/s

Working Power Mode: 7 W - 15 W adjustable power consumption mode

AI Computing Performance: Up to 40 sparse INT8 TOPS / 20 dense INT8 TOPS

Storage Support: microSD card + external NVMe solid-state drive

Core Computing Tasks: LiDAR point cloud data processing, SLAM mapping, real-time vehicle localization, global path planning, local obstacle avoidance, vehicle longitudinal and lateral closed-loop control, emergency brake logic judgment. All tasks run locally without remote computing assistance.

7. Autonomous Driving Software System Overview

7.1 Software Overall Architecture

The whole software system is developed based on Robot Operating System (ROS 2), divided into four core modules: perception module, localization and mapping module, planning module and vehicle control module. All algorithm programs run independently on the on-board Jetson Orin Nano, with no external data receiving channel during the whole race.

7.2 Core Software Module Introduction

Perception Module: The perception module receives raw point cloud data from the Hokuyo UST-10LX 2D LiDAR and processes the data in real time on the onboard main computing unit. The system first pre-processes the raw point cloud by removing invalid points that are too close, too far away, or located behind the vehicle. Only the local point cloud in front of the vehicle that is relevant to driving decisions is retained. A grid-based density filtering method is then applied to reduce the influence of sparse noise points and occasional LiDAR measurement errors.

After point cloud filtering, the perception module identifies obstacles, track boundaries, and local drivable areas based on the spatial distribution of the LiDAR points. The system analyzes the point cloud distribution on both sides of the vehicle and evaluates the spatial gaps between point clusters to estimate the left and right boundaries of the local driving corridor as well as its center direction. This method provides real-time environmental information for path planning and vehicle control without relying on cameras, GPS, or external computing devices. No onboard intelligent recognition chip is used; all perception algorithms are executed by the main computing unit.

Mapping & Localization Module: The mapping and localization module is responsible for track map construction and real-time vehicle pose estimation. During the group mapping stage, the vehicle performs LiDAR-based SLAM mapping using 2D LiDAR data to build a unified map of the track environment. The system also supports official map files in png, yaml, and pbsstream formats. Based on the generated or provided map, the system can obtain track boundaries, drivable areas, and the basic environmental information required for global reference trajectory generation.

During autonomous driving, the vehicle adopts LiDAR-based map matching localization. The

real-time LiDAR scan is matched with the pre-built or officially provided track map to estimate the vehicle position and heading in the map coordinate system. The localization result is then provided to the path planning and vehicle control modules for reference trajectory tracking and closed-loop vehicle control. This localization method does not rely on VIO, GPS, or other auxiliary positioning methods.

Path Planning Module: The path planning module provides two available methods: global path planning based on the SLAM mapping result, and local path planning based on real-time 2D LiDAR point cloud data. Both methods are deployed on the onboard computing platform and can be selected according to the race strategy, track condition, and running mode.

In the global path planning method, the system uses the SLAM mapping result to obtain track boundaries, drivable areas, and the overall track layout, and then generates a full-course reference trajectory based on the map. This reference trajectory provides a stable global driving target for the vehicle and is suitable for high-speed continuous lap driving in stable track environments with complete map information.

In the local path planning method, the system directly estimates the drivable area in front of the vehicle based on real-time LiDAR point cloud data. Through point cloud filtering, boundary identification, and drivable free-space searching, the vehicle can select a safe local driving direction in real time, allowing it to handle obstacle avoidance, boundary changes, and local environmental disturbances.

These two planning methods support global trajectory following and local real-time obstacle avoidance under different track conditions. All planning computations are completed by the onboard main computing unit, without external computing devices or real-time manual intervention.

Vehicle Control Module: The vehicle control module provides two available control methods: trajectory tracking control based on the global reference trajectory, and real-time reactive control based on the local planning result. Both methods are deployed on the onboard computing platform and can be selected according to the path planning method, track condition, and running mode.

In the global reference trajectory tracking method, the system matches the real-time vehicle pose from the localization module with the full-course reference trajectory generated from the SLAM map. It calculates the lateral error and heading error of the vehicle relative to the reference trajectory, and uses a PID controller to generate the corresponding steering angle command. This method is mainly used for stable and high-speed continuous lap driving along a fixed reference trajectory.

In the local-planning-based control method, the system directly generates steering commands according to the local drivable corridor and target driving direction obtained from real-time LiDAR point cloud data. When obstacles, boundary changes, or narrow passages appear ahead, the controller adjusts the steering command in real time to guide the vehicle toward a safer drivable area.

Finally, the control module sends steering and throttle commands to the chassis actuators and matches the power output of the motor and ESC, realizing real-time closed-loop control of vehicle speed and steering angle. All control computations are completed by the onboard main computing unit, without external computing devices or real-time manual intervention.

7.3 Human Intervention Restriction Description

No keyboard, mouse or joystick manual intervention during the whole qualification and head-to-head race process;

No wireless data transmission to the vehicle during normal autonomous driving;

Manual emergency brake will not be actively triggered during normal racing process;

Only one laptop is allowed to connect to the vehicle for offline visualization and debugging, no real-time remote control command output.

8. Adaptability Analysis for Competition Track Environment

The competition track is built on smooth concrete ground with urethane coating, featuring low road friction coefficient. This vehicle has completed targeted optimization aiming at this environment:

Optimize speed control algorithm to avoid tire slipping under low friction condition;

Adjust LiDAR ground filtering parameters to adapt to smooth ground reflection characteristics;

Optimize steering response parameters to improve vehicle stability during high-speed cornering;

The vehicle overall weight and chassis ground clearance are matched with indoor 8m×22m small-size track, meeting high-speed continuous lap driving demand.

9. Full Vehicle Rule Compliance Summary

All configurations of the autonomous racing vehicle fully comply with all hardware constraints, software inspection requirements and race operation specifications of ROBORACER GRAND PRIX 2026. The key compliance items are summarized as follows:

Chassis dimension meets official width and length limits;

Maximum single battery 4S, no over-limit power battery;

Single LiDAR equipped, no prohibited VIO camera;

Single motor power system, complying with powertrain rules;

Exclusive single on-board computing unit, all calculation finished locally;

Equipped with qualified detection box and soft foam bumper;

Toggle-type emergency brake system meets software inspection standard;

No manual intervention and external data transmission during races;

No prohibited indoor GPS and other auxiliary sensors.

10. Declaration

Our team declares that all hardware and software configurations described in this document are consistent with the actual vehicle. No unreported modified parts or rule-violating devices are installed. Our team will accept all official hardware and software inspections arranged by the competition organizing committee, and abide by all race rules and on-site referee judgments during the whole competition.